ACE MICRO

500g Sumobot



What is ACE Micro?

ACE Micro V2 is the second version of my mini sumobot in the ACE lineup. It features custom neon green silicone wheels that I made using two-part silicone. Version 1 had pink wheels, but mid-year, my taste shifted and I thought, "Why not neon so it stands out?"—especially so I could keep track of it during fast-paced battles on the dohyo ring. I chose green because it represents the color of my school, robotics organization, and mathematics organization.

Why ACE?

Friends initially thought the name ACE came from my surname, ACEna. That was a creative connection, but I feel there is still a gap between me and "ace" that I must never stop working to bridge. I specifically chose the word "ace" to build a sustainable brand for the lineup—one that embodies the phrase "keep acing," which I always share with my juniors at Animo SPEAR.

What was the Version 1?

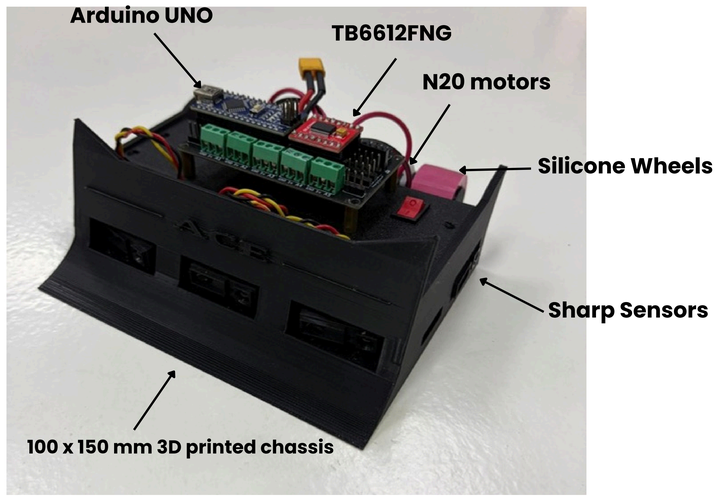


Figure 1. Version 1 of ACE Micro

V1 weighed about 290 g and ran N20 motors, so it was built for speed and had no push. Opponents moved it easily. Its Sharp sensors measure distance, so it could not see the edge line and drove itself out of the ring, which gives the opponent a point. V2 fixes this with JGA25-370 motors, more weight, and two line sensors under the base.



Figure 2. ACE Micro V2 CAD Model

I modeled V2 with a friend on a desktop in my school's library during our free hours. It took 3 prints. The small leftover print is now the start of ACE Nano, a compact sumobot my juniors and peers can use to test their sumobot code before running it for real.

This was my third time creating silicone wheels. Drawing from my experience with the previous two pairs, I reduced the ratio of Part B (hardener) to make the silicone slightly "tacky" and increase traction.



Figure 3. The creation of silicone wheels

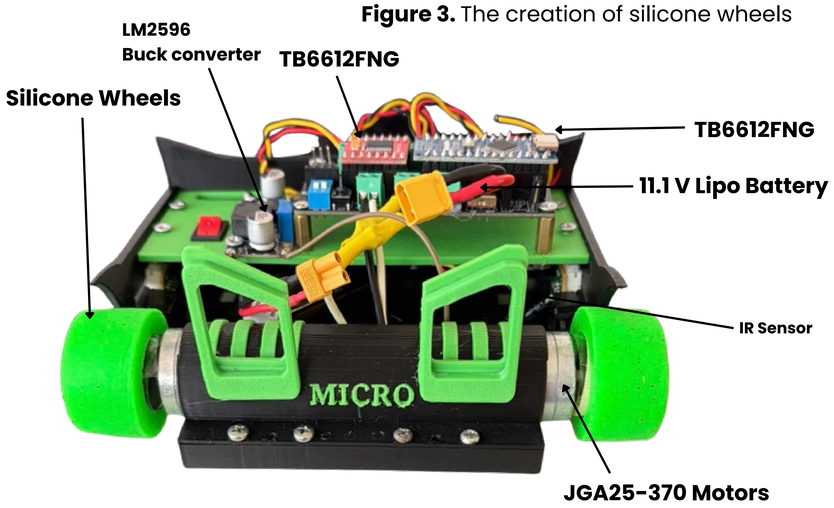


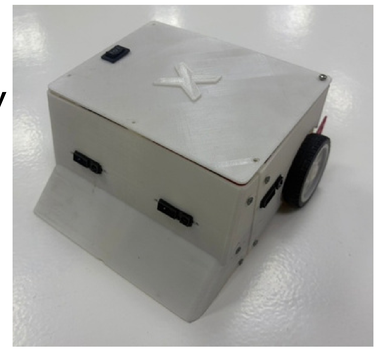
Figure 4. The rear of ACE Micro

The Nano's 5V regulator was overloaded by the sensors. I lost one Nano, and I think the regulator chip broke. A buck converter now gives the Sharp and line sensors their own 5V rail.

The parts and the upgrade

Version 2 features a 150x150 mm 3D-printed chassis. The N20 motors were upgraded to JGA25-370 620 RPM motors to add both torque and weight to the previously underweight Version 1 (~290 g). I added a buck converter to create a separate 5V power rail for the Sharp distance sensors and line sensors, preventing the Arduino Nano's built-in 5V regulator from overloading. Line sensors were also installed on the bottom-left and bottom-right of the base to keep the sumobot from driving off the dohyo ring. Powering the system is a 3S LiPo battery connected to a high-current motor driver, ensuring rapid acceleration and sufficient pushing power during clashes.

Where it all started



I wasn't originally a "sumobot person"—I started with line-following competitions. However, I developed a strong interest in sumobots after trying out a basic kit in our robotics laboratory and being influenced by my senior, Kuya Zen.

Figure 5. The Erovoutika Sumobot kit

The green handle at the back is there because other teams used zipties to lift malfunctioning robots out of the ring. I added a proper carry handle.

ACE MICRO: VERSION 1 TO VERSION 2

	Version 1	Version 2
Motors	N20 - 20,000 RPM	JGA25-370 - 620 RPM
Weight	290g, underweight	Exactly 500g, fits the weight class and more push
Edge detection	None	2 IR Line Sensors underneath
Power	Nano 5V regulator only (overloaded)	Buck Converter for own 5V Rail
Wheels	Pink, not tacky for sumo bot	Neon Green, tacky enough
Handle	None	Green carry handle

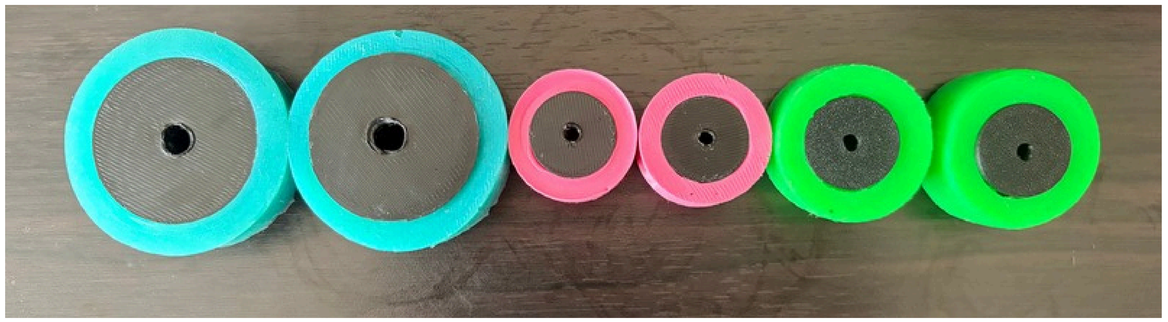


Figure 1. DIY Silicone wheels: Mega attempt 1 (blue), Micro V1 (pink), Micro V2 attempt 3 (neon green).

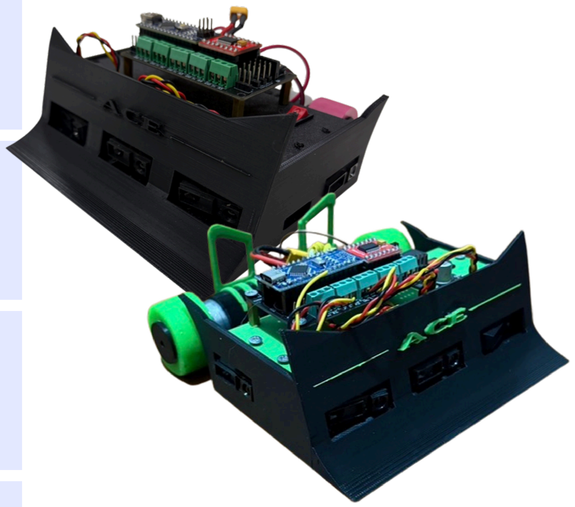


Figure 2. V1 and V2 comparison

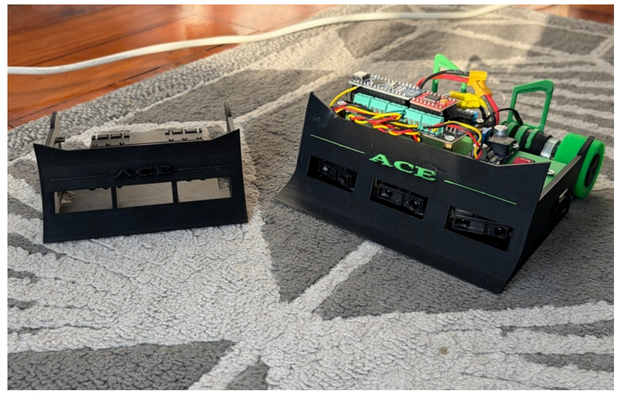


Figure 3. The V2 beside the failed chassis print that will become ACE Nano

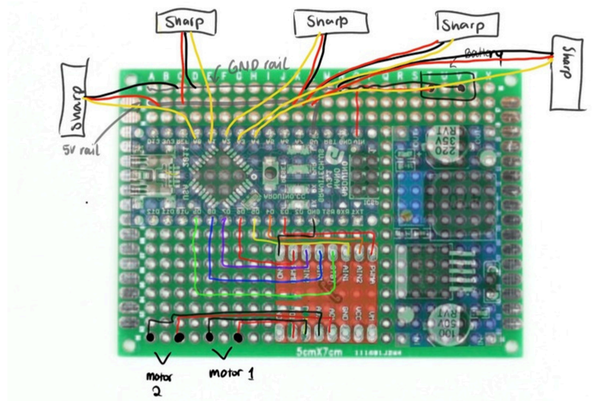


Figure 4. The layout planning

What I learned: V1 was fast but had no push, and it could not see the ring edge. More weight, tacky wheels, and line sensors fixed that, and a separate power rail stopped the Nano from overloading.